Lecture 8

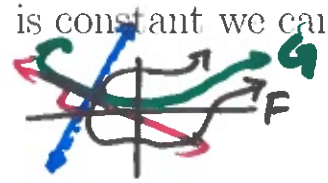
Applications of 1st-Order Equations

In this lecture, we will explore various applications to geometry and physics.

Orthogonal Trajectories

A problem that arises in many different disciplines is that of finding a family of curves that is orthogonal (perpendicular) to a given family of curves. One basic geometric idea is given a family of level curves $f(x, y) = k$ the perpendicular flow across the level curves (moving from one curve to another) is given by another family of orthogonal curves called an **orthogonal trajectory**.

This problem arises in electromagnetic theory when trying to determine the equipotential flux given the lines of an electric field. In the theory of heat conduction, given curves or lines of constant temperature (called isotherms) the orthogonal trajectories given the curves of heat flow. In fluid mechanics, given the curves along which the fluid potential is constant we can determine the fluid flow.



$F \perp G$

A family of curves is given by some equation $F(x, y) = k$. We can find the slope of any curve by using the total differential

$$dF = \frac{\partial F}{\partial x}dx + \frac{\partial F}{\partial y}dy = 0 \quad \Rightarrow \boxed{\frac{dy}{dx} = -\frac{\partial F/\partial x}{\partial F/\partial y}}$$

tangent line slope to $F(x,y)=k$

When two curves are orthogonal the slopes of their tangent lines must be

$$\frac{\partial F}{\partial x}dx + \frac{\partial F}{\partial y}dy = 0 \Rightarrow \frac{\partial F}{\partial y}dy = -\frac{\partial F}{\partial x}dx$$

$$\Rightarrow \frac{dy}{dx} = -\frac{\frac{\partial F}{\partial x}}{\frac{\partial F}{\partial y}}$$

the negative reciprocal of each other. Therefore, the slopes of the orthogonal trajectories to $F(x, y) = k$ must satisfy

$$\frac{dy}{dx} = + \frac{\frac{\partial F}{\partial y}}{\frac{\partial F}{\partial x}}$$

which translates to the equation

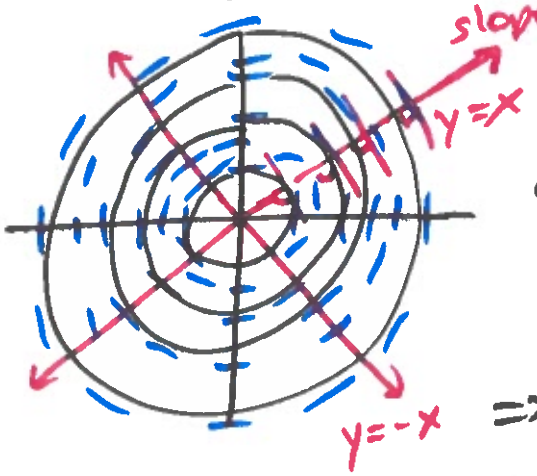
$$\frac{\partial F}{\partial x} \cdot \frac{dy}{dx} = \frac{\partial F}{\partial y}$$

$$\Rightarrow \frac{\partial F}{\partial x} dy = \frac{\partial F}{\partial y} dx$$

$$\Rightarrow \boxed{\frac{\partial F}{\partial y} dx - \frac{\partial F}{\partial x} dy = 0}$$

orthogonal trajectory of $F = k$

Example: Find the family of orthogonal trajectories to the family of circles $x^2 + y^2 = k$



$F(x,y) = k$

$\Rightarrow F(x,y) = x^2 + y^2 \rightarrow \frac{dy}{dx} = -\frac{x}{y}$ circles

orthogonal trajectories:

$$\frac{\partial F}{\partial y} dx - \frac{\partial F}{\partial x} dy = 0$$

$\Rightarrow 2y\, dx - 2x\, dy = 0$ separable

$\Rightarrow 2y\, dx = 2x\, dy$

$\Rightarrow \int \frac{1}{x} dx = \int \frac{1}{y} dy$

$\Rightarrow \ln|x| + C = \ln|y|$

$\Rightarrow e^{\ln|x| + C} = y$

$\Rightarrow \boxed{y = Cx}$ lines through $(0,0)$ of slope C.

Example: Streamlines associated with a certain fluid flow can be represented by the family

$$y = x - 1 + ke^{-x}$$

Find the velocity potentials of the flow by determining the orthogonal trajectories of the streamlines.

write in the form. $F(x,y)=k$

$$y = x-1+ke^{-x}$$

$$\Rightarrow y-x+1 = ke^{-x} \Rightarrow \underbrace{e^x(y-x+1)}_{F(x,y)} = k$$

orthogonal trajectories: $F_y\,dx - F_x\,dy = 0$

$$\underset{F_y}{e^x}\,dx - (e^x(y-x+\cancel{1}) + e^x(\cancel{-1}))\,dy = 0$$

$$\Rightarrow e^x dx - (e^x y - xe^x)\,dy = 0$$

$(e^x > 0)$ divide by $e^x \Rightarrow 1\cdot dx - (y-x)\,dy = 0$

$$\Rightarrow dx = (y-x)\,dy$$

$$\Rightarrow 1 = (y-x)\frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{1}{y-x}$$

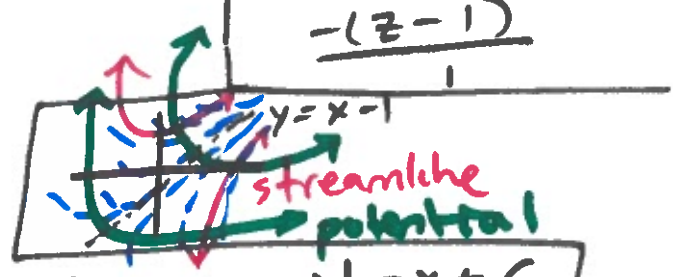
Substitution $\frac{dy}{dx} = G(-x+y+0)$, $z = y-x$

$$\frac{dz}{dx} = \frac{dy}{dx} - 1 \Rightarrow \frac{dy}{dx} = \frac{dz}{dx} + 1$$

$$\Rightarrow \frac{dz}{dx} + 1 = \frac{1}{z}$$

$$\Rightarrow \frac{dz}{dx} = \frac{1-z}{z} \Rightarrow \int \frac{z}{1-z}\,dz = \int dx$$

$$\begin{array}{r} -1 \\ 1-z\,\overline{)\,z} \\ -(z-1) \\ \hline 1 \end{array}$$



$$= \int -1 + \frac{1}{1-z}\,dz = x + C$$

$$= -z - \ln|1-z| = x + C \Rightarrow -(y-x) - \ln|1-(y-x)| = x + C$$

implicit.

Exponential Growth and Decay

Suppose we have some amount of "something" (be it money, population, mass, etc) and the change in the amount is proportional to the current amount. Such a situation arises in banking and finance, radioactive decay, population growth, artifact dating, etc. We can write this as a 1st-order equation

$$\frac{dP}{dt} = kP$$

This is a separable equation. If we solve it we obtain

divide by P
multiply by dt

$$\int \frac{1}{P}\,dP = \int k\,dt$$

$$\Rightarrow \ln|P| = kt + C$$

$$\Rightarrow P = e^{kt+C} = Ce^{kt}$$

$P(0) = P_0$ starting amount

$$P_0 = Ce^{k(0)} = C$$

$$P(t) = P_0 e^{kt}$$

growth $k>0$
decay $k<0$

Example: You start with 1000 bacteria. The number of bacteria doubles every twenty minutes. How many bacteria are there in 48 hours?

$P(t) = P_0 e^{kt}$, $P(0) = 1000$, $P(1)$ = pop after 20 minutes

48 hours = 144 sets of 20 minutes.

$P(t) = 1000e^{kt}$?

$P(1) = 2000 = 1000e^{k(1)} \Rightarrow e^k = 2 \Rightarrow \boxed{k = \ln(2)}$

$P(t) = 1000e^{\ln(2)t} = 1000 \cdot 2^t$

Amount of bacteria in 48 hours: $P(144) = 1000 \cdot 2^{144}$

weight of one E. Coli bacterium $\approx 0.85 \times 10^{-15}$ kg | $\approx 2.23 \times 10^{46}$ bacteria

Total mass $\approx 1.89 \times 10^{31}$ kg earth: 5.97×10^{24} kg.

How can we fix this model?

Logistic Models

In order to account for unbounded exponential growth, we can adjust the model to slow the rate of growth the closer the population is to some maximum. This was explored by Verhurst around 1936. The model that was developed is called a **logistic model** and it is given by the equation

$$\frac{dP}{dt} = rP\left(1 - \frac{P}{K}\right)$$

where r is a growth parameter and K is the **carrying capacity**.

This is a separable equation which can be separated as

$$\frac{dP}{P\left(1 - \frac{P}{K}\right)} = r\,dt \quad \Rightarrow \quad \boxed{P(t) = \frac{P_0 K}{P_0 + (K - P_0)e^{-rt}}}$$

where P_0 is the initial population.

Example: Suppose an initial population of bacteria is 3mg with a carrying capacity of 100mg and growth parameter 0.2/hr.

$$\boxed{\frac{dP}{dt} = 0.2P\left(1 - \frac{P}{100}\right)}$$

So $P(t) = \dfrac{3 \cdot 100}{3 + (100 - 3)e^{-0.2t}}$

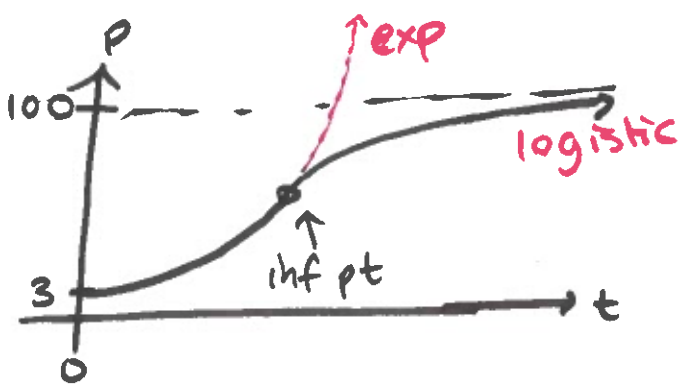
$$= \frac{300}{3 + 97e^{-0.2t}}$$

$$\lim_{t \to \infty} P(t)$$

$$\lim_{t \to \infty} \frac{300}{3 + 97\cancelto{0}{e^{-0.2t}}} = \frac{300}{3} = 100$$

K

Graph



Newton's Law of Cooling

Newton's Law of Cooling is given by a 1st-order equation that relates the change in temperature of an object with relation to the temperature of the environment it is in. It is given by

$T' \sim k(T_e - T)$

$$\boxed{\frac{dT}{dt} = k(T_e - T)}$$

where $k > 0$ is a constant for the rate of cooling, T is the temperature at time t, T_0 is the initial temperature of the object, and T_e is the surrounding (environmental) temperature.

ambient

This equation is separable:

$$\frac{dT}{T_e - T} = kdt \quad \Rightarrow \quad \ln|T_e - T| = kt + C \quad \Rightarrow \quad T_e - T = e^{-kt+C}$$

Solving for T we obtain

$T = T_e + Ce^{-kt}$ ✓ $\quad T_0 = T_e + Ce^0$

and using the initial condition that $T(0) = T_0$ we see that $\boxed{C = T_0 - T_e.}$

Therefore Newton's Law of Cooling is given by

$$\boxed{T(t) = T_e + (T_0 - T_e)e^{-kt}}$$

Example: A cup of hot coffee is initially 95°C and cools to 80°C in 5 minutes while sitting in a room of temperature 21°C. How long will it take for the coffee to reach 50°C?

$T_0 = T(0) = 95$, $T_e = 21$

$T(5) = 80$

t? $T(t) = 50$?

$$T(t) = 21 + (95 - 21)e^{-kt}$$

$$\Rightarrow T(t) = 21 + 74e^{-kt}$$

$T(5) = 80$

$$\Rightarrow 80 = T(5) = 21 + 74e^{-k(5)}$$

$$\Rightarrow 59 = 74e^{-5k}$$

$$\Rightarrow e^{-5k} = \frac{59}{74} \Rightarrow -5k = \ln\left(\frac{59}{74}\right)$$

$$\Rightarrow k = -\frac{1}{5}\ln\left(\frac{59}{74}\right)$$

$$\approx 0.045306$$

so $50 = T(t) = 21 + 74e^{\underbrace{-0.045306}_{k}t}$

$$\Rightarrow 29 = 74e^{-0.045306t}$$

$$\Rightarrow e^{-0.045306t} = \frac{29}{74} \Rightarrow -0.045306t = \ln\left(\frac{29}{74}\right)$$

$$\Rightarrow t = \frac{-\ln\left(\frac{29}{74}\right)}{0.045306} \approx \underline{20.7 \text{ minutes}}.$$